Epsilon-Approximate Credal Variable Elimination: A Fully Polynomial-Time Approximation Scheme

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Abstract

The exact Credal Variable Elimination (CVE) algorithm may produce intermediate potentials whose cardinality grows exponentially. By relaxing the pruning criterion to allow *epsilon-dominated* functions to be removed, we obtain an approximate algorithm whose error is bounded by a user-specified tolerance $\varepsilon > 0$. This epsilon-approximate CVE constitutes a fully polynomial-time approximation scheme (FPTAS) for marginal inference in credal networks of bounded treewidth and bounded variable cardinality.

1 Motivation

In exact CVE (see companion document `credal_ve.tex`), the pruning step removes only strictly dominated functions: f is pruned if there exists g such that $g(\mathbf{y}) \geq f(\mathbf{y})$ for all $\mathbf{y}$, with strict inequality somewhere. While this preserves exact bounds, the number of surviving functions can grow exponentially across elimination steps.

The key insight of Mauá et al. (2012) is to relax the dominance criterion: remove functions that are *almost* dominated, controlled by a tolerance parameter ε. This trades bound tightness for computational efficiency.

2 Epsilon-Dominance

Definition 1 (ε-Dominance)**.** *Let $\varepsilon \geq 0$. A function f is ε-**dominated** by function g if:*

$$g(\mathbf{y}) \geq f(\mathbf{y}) - \varepsilon \quad \forall \mathbf{y}. \tag{1}$$

When $\varepsilon = 0$, this reduces to standard dominance. For $\varepsilon > 0$, more functions are classified as dominated and thus pruned, reducing the potential's cardinality.

Definition 2 (ε-Pruning)**.** *The ε-pruning of a potential ϕ is:*

$$\text{prune}_\varepsilon(\phi) = \{f \in \phi : \nexists g \in \phi \setminus \{f\} \ \textit{s.t.}\ g(\mathbf{y}) \geq f(\mathbf{y}) - \varepsilon \, \forall \mathbf{y}\}.$$

3 Algorithm

The algorithm is identical to exact CVE (Algorithm **??**) except that the pruning step at line **??** uses ε-pruning instead of exact pruning.

Algorithm 1 Epsilon-Approximate Credal Variable Elimination

Require: Extreme points $\text{ext}\, K(X_i \mid \text{Pa}(X_i))$ for each node, query Z, evidence $\mathbf{Y} = \mathbf{y}$, tolerance $\varepsilon > 0$
Ensure: Approximate bounds $[\underline{P}_\varepsilon,\ \overline{P}_\varepsilon]$

Initialize potentials Γ from extreme points and evidence indicators ▷ Same as exact CVE
Compute elimination ordering $\mathbf{o}$ excluding Z
for each X_i in ordering $\mathbf{o}$ **do**
  $\Gamma_{X_i} \leftarrow \{\phi \in \Gamma : X_i \in \text{scope}(\phi)\}$
  $\Gamma \leftarrow \Gamma \setminus \Gamma_{X_i}$
  $\psi \leftarrow \prod\{\phi \in \Gamma_{X_i}\}$
  $\psi' \leftarrow \sum_{X_i} \psi$
  $\psi'' \leftarrow \text{prune}_\varepsilon(\psi')$ ▷ ε-pruning instead of exact pruning
  $\Gamma \leftarrow \Gamma \cup \{\psi''\}$
end for
Extract bounds from remaining potentials ▷ Same as exact CVE
return $[\underline{P}_\varepsilon,\ \overline{P}_\varepsilon]$

3.1 Epsilon-Pruning Procedure

Algorithm 2 ε-Prune

Require: Potential $\phi = \{f_1, \ldots, f_m\}$, tolerance ε
Ensure: Pruned potential ϕ'

dominated$[i] \leftarrow$ false for $i = 1, \ldots, m$
for $i = 1, \ldots, m$ **do**
  if dominated$[i]$ **then continue**
  end if
  for $j = 1, \ldots, m,\ j \neq i$ **do**
    if dominated$[j]$ **then continue**
    end if
    if $f_j(\mathbf{y}) \geq f_i(\mathbf{y}) - \varepsilon$ for all $\mathbf{y}$ **then**
      dominated$[i] \leftarrow$ true
      break
    end if
  end for
end for
return $\phi' = \{f_i : \neg\,\text{dominated}[i]\}$

4 Theoretical Properties

Theorem 1 (FPTAS, Mauá et al. 2012)**.** *For credal networks of bounded treewidth w^* and bounded variable cardinality k, the ε-approximate CVE runs in time polynomial in the input size and in $1/\varepsilon$, and produces bounds within ε of the exact bounds:*

$$\underline{P}(z \mid \mathbf{y}) \leq \underline{P}_\varepsilon(z \mid \mathbf{y}) \leq \underline{P}(z \mid \mathbf{y}) + O(\varepsilon),$$

and similarly for the upper bound.

Bound direction. Since ε-pruning only removes functions (never adds them), the approximate algorithm produces bounds that are at least as wide as the exact bounds:

$$\underline{P}_\varepsilon \leq \underline{P}, \qquad \overline{P}_\varepsilon \geq \overline{P}.$$

Thus the ε-approximate bounds are always valid *outer* approximations.

5 Running Example

Example 1 (Alarm Network)**.** *We use the same alarm network as in* `credal_ve.tex` *with variables* B*,* E*,* A*, and* CD*.*

Query: $P(B{=}1)$**, elimination order:** E, A, CD**.**

ε	$\underline{P}(B{=}1)$	$\overline{P}(B{=}1)$	*Time (s)*
0 *(exact)*	*0.1000*	*0.2000*	*3.69*
0.001	*0.1000*	*0.2000*	*3.14*
0.01	*0.1000*	*0.2000*	*1.53*
0.1	*0.1000*	*0.2000*	*0.009*

For this query, even aggressive pruning ($\varepsilon = 0.1$*) produces the same bounds because* B *is a root node with only two vertices. The speedup comes from pruning functions in the intermediate potentials during elimination of* E*,* A*, and* CD*.*

Query: $P(A{=}1 \mid B{=}0, E{=}0)$**.**

ε	$\underline{P}(A{=}1 \mid \cdot)$	$\overline{P}(A{=}1 \mid \cdot)$	*Time (s)*
0 *(exact)*	*0.0000*	*0.0444*	*0.012*
0.01	*0.0000*	*0.0444*	*0.002*
0.1	*0.0000*	*0.0000*	*0.001*

With $\varepsilon = 0.1$*, the interval collapses to* $[0, 0]$ *because the true upper bound (*0.044*) is smaller than* ε*, so the function achieving the maximum gets* ε*-dominated. This illustrates the tradeoff: large* ε *gives speed but may lose precision when probabilities are small relative to* ε*.*

6 Practical Guidance

- Start with $\varepsilon = 0$ (exact) for small networks.
- For larger networks, use $\varepsilon = 10^{-3}$ to 10^{-2} as a good balance between speed and accuracy.
- Avoid ε larger than the smallest probability of interest.
- The algorithm is most effective when the network treewidth is moderate (the intermediate potentials are the bottleneck, and pruning directly controls their size).

7 References

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2. D.D. Mauá, F.G. Cozman. Thirty years of credal networks: Specification, algorithms and complexity. *IJAR*, 126:133–157, 2020.

3. R. Marinescu et al. Credal Marginal MAP. *NeurIPS*, 2023.